

Course Level: Undergraduate Open Elective (B. Tech)

BL: 2 Understand

CO: Explain the fundamental principles of nutrition.

Module 2: Digestion & Absorption Patterns

1. General Principles

Nutrient absorption occurs mainly in the small intestine.

Each macronutrient (carbohydrates, proteins, fats) has distinct digestion and absorption pathways.

Transport processes are highly specific and involve specialized enzymes and membrane transporters.

2. Carbohydrate Absorption

Digestion steps:

Intraluminal hydrolysis: polysaccharides → disaccharides.

Membrane hydrolysis: brush-border disaccharidases → monosaccharides.

Absorption:

Only monosaccharides (glucose, galactose, fructose) are absorbed.

Transport occurs in two steps:

Uptake across apical membrane.

Exit across basolateral membrane.

Transporters:

Glucose & galactose: SGLT1 (Na⁺-dependent active transport).

Fructose: GLUT5 (facilitated diffusion).

All monosaccharides exit via GLUT2.

3. Protein Absorption

Digestion pathways:

Luminal proteases → amino acids.

Luminal proteases → oligopeptides → brush border peptidases → amino acids.

Oligopeptides taken up → cytosolic enzymes → amino acids → blood.

Oligopeptides taken up → direct transport to blood.

Efficiency: >96% absorbed; <4% excreted in stool.

Sources:

Dietary proteins.

Endogenous proteins (enzymes, hormones, immunoglobulins, epithelial cells).

Factors influencing digestion:

Amino acid composition (proline-rich proteins less digestible).

Food source (animal proteins > plant proteins).

Processing (cooking, dehydration reduce digestibility).

4. Preparing for Protein Absorption

Proenzymes: secreted in inactive form to prevent auto-digestion.

Pepsinogen → pepsin.

Trypsinogen → trypsin.

Chymotrypsinogen → chymotrypsin.

Proelastase → elastase.

Procarboxypeptidase A/B → carboxypeptidases.

Contribution: Pancreatic proteases convert ~70% proteins → oligopeptides, ~30% → free amino acids.

5. Brush Border & Cytoplasmic Digestion

Brush border peptidases: act on oligopeptides (3–8 AA).

Cytoplasmic peptidases: hydrolyze dipeptides & tripeptides.

Enzyme diversity matches peptide bond specificity.

6. Absorption of Intact Proteins

Neonates: absorb intact proteins (passive immunity transfer).

Adults: limited absorption of intact proteins via:

Enterocytes.

M cells.

Role: mucosal immunity, potential disease processes.

7. Oligopeptide Absorption

Uptake via H⁺-gradient-driven transporters (different from amino acid transporters).

Intracellular peptidases hydrolyze absorbed oligopeptides → amino acids.

8. Amino Acid Absorption

Enterocytes use ~10% of absorbed amino acids for protein synthesis.

Basolateral transport is bidirectional:

3 transporters mediate exit into blood

2 transporters mediate uptake from blood for cell nutrition.